CHOCOLATE & BEVERAGE SALES, AND PROFITABILITY ANALYSIS

Objective:

The objective of this project is to analyze chocolate and beverage sales data to evaluate business performance in terms of revenue, expenses, and profitability across different countries, states, and products. The project aims to identify high-performing and low-performing chocolate and beverage products, understand regional sales patterns, and analyze profit margins to support data-driven business decisions.

Dataset Description:

The dataset consists of an order-level sales fact table along with supporting dimension tables such as Product Dimension and Salesperson Dimension. The sales fact table includes attributes like Order ID, Product ID, Sales ID, Country, State, Order Date, Sales Amount, Expenses, Selling Price, Product Cost, Profit, Profit Margin, Order Status, Customer Count, and box size distribution (Small, Medium, Large).

The dimension tables provide additional descriptive information related to products and salespersons, enabling detailed analysis of sales performance across product categories, regions, and sales representatives.

Problem Statements:

- The business lacks a centralized system to track and monitor chocolate and beverage sales performance across products, countries, categories, and salespersons.
- It is difficult to identify which products generate the highest sales and profit, and which products underperform.
- The company has limited visibility into country-wise and state-wise sales performance, making it challenging to identify high-performing and low-performing regions.
- There is no clear understanding of seasonal sales patterns and monthly demand fluctuations.
- High cancellation rates in certain countries and for certain products are causing revenue loss and affecting customer satisfaction.
- The business is unable to easily determine which product categories contribute the most to overall profit.
- There is limited insight into salesperson performance, making it difficult to evaluate individual contribution to sales and cancellations.
- The company faces challenges in inventory planning and demand forecasting due to the lack of historical sales trend analysis.
- Operational issues such as supply chain delays, stock shortages, and delivery problems are difficult to identify without proper cancellation analysis.
- There is a need for an interactive dashboard that provides actionable insights into sales, profit, cancellations, regional performance, and product demand to support better business decision-making.

Sales Table - Column Description

Column Name	Description
Order_ID	A unique identifier assigned to each customer order. Used to track and reference individual orders.
Product_ID	A unique identifier assigned to each product.
SalesPerson_ID	A unique identifier assigned to each salesperson in the organization. Used to track individual sales performance and link sales transactions to specific sales personnel.
Country	Name of the country where the sale was made.
State	Name of the state or region within the country.
Date	Date on which the order was placed or sales transaction occurred.
Selling Price	Price at which the product was sold per unit or box.
Product Cost	Cost incurred to produce or procure the product.
Order Status	Status of the order (Delivered, Cancelled, etc.).
Customers	Number of customers associated with the order or sales record.
Quantity	The total number of units of a product that have been sold.
Sales	Total revenue generated from the sale of products.
Expenses	Total expenses incurred for the order, including operational and distribution costs.
Profit	The absolute monetary gain from a sale activity. Calculated as **Revenue – Cost**. Represents the actual earnings.
ProfitMargin	The relative profitability expressed as a percentage. Helps evaluate how efficiently revenue is converted into profit.

Product-Dim Table Description:

Column Name	Description
Product_ID	A unique identifier assigned to each product.
Product_Name	The name of the product for easy identification.
Product_Category	The classification or type of the product

SalesPersion_Dim Table Description:

Column Name	Description
SalesPerson_ID	A unique identifier assigned to each salesperson in the organization. Used to track individual sales performance and link sales transactions to specific sales personnel.
Sales_Person_Name	The full name of the salesperson for identification purposes.

Data Import Process

The dataset was imported into the analysis environment using **Power BI Desktop**. The following steps were followed to load the data:

1. Navigated to the **Home** tab in Power BI Desktop.
2. Selected **Get Data** and chose **From Excel Workbook**.
3. Browsed and selected the *Chocolate_Sales_Dataset* file.
4. Chose the required worksheets from the workbook.
5. Clicked **Transform Data**, and the necessary data cleaning steps were performed in **Power Query** before loading the data into the model for further analysis.

Data Cleaning & Transformation:

- **ID Standardized**
  - Data preprocessing was performed in Power Query by applying the **Clean** and **Trim** functions on the **Order ID, Product ID, and Salesperson ID** fields to remove hidden characters and unnecessary whitespace, thereby ensuring consistent formatting and preventing key mismatches during data modeling.
- **Country Standardized**
  - In Power Query, the **Country** columns was standardized by applying the **Clean** and **Trim** functions to remove unwanted spaces and hidden characters, followed by the **Capitalize Each Word** transformation to ensure consistent and proper country name formatting.
- **Sate Standardized**
  - In Power Query, the **Sate** columns was standardized by applying the **Clean** and **Trim** functions to remove unwanted spaces and hidden characters, followed by the **Capitalize Each Word** transformation to ensure consistent and proper country name formatting.

- **Date Standardized**
  - The **Date** column was standardized in Power Query by applying the **Clean** and **Trim** transformations to remove unwanted spaces and hidden characters, and its data type was converted to **Date format** to enable accurate time-based analysis.
- **Sales Standardized**
  - The Sales columns were standardized in Power Query by applying the **Clean** and **Trim** transformations to remove unwanted spaces and hidden characters, and their data types were converted to **Fixed Decimal** format to ensure accurate numerical analysis.
- **Expenses Standardized**
  - The Expenses columns were standardized in Power Query by applying the **Clean** and **Trim** transformations to remove unwanted spaces and hidden characters, and their data types were converted to **Fixed Decimal** format to ensure accurate numerical analysis.
- **Selling Price Standardized**
  - The Selling Price columns were standardized in Power Query by applying the **Clean** and **Trim** transformations to remove unwanted spaces and hidden characters, and their data types were converted to **Fixed Decimal** format to ensure accurate numerical analysis.
- **Boxes Standardized**
  - The Boxes columns were standardized in Power Query by applying the **Clean** and **Trim** transformations to remove unwanted spaces and hidden characters, and their data types were converted to **Whole Number** to ensure accurate numerical analysis.
- **Product Cost Standardized**
  - The Product Cost columns were standardized in Power Query by applying the **Clean** and **Trim** transformations to remove unwanted spaces and hidden characters, and their data types were converted to **Whole Number** to ensure accurate numerical analysis.
- **Order Status Standardized**
  - The Order Status columns were standardized in Power Query by applying the **Clean** and **Trim** transformations to remove unwanted spaces and hidden characters, and their data types were converted to **Text**.
- **Customers Standardized**
  - The Customers columns were standardized in Power Query by applying the **Clean** and **Trim** transformations to remove unwanted spaces and hidden characters, and their data types were converted to Whole Number to ensure accurate numerical analysis.

- **Profit Standardized**
  - A custom column called Profit was created using a DAX formula to calculate row-level profit as Sales minus Expenses. Data cleansing was performed using Clean and Trim transformations to remove unwanted spaces and hidden characters. All monetary columns were converted to Fixed Decimal Number data type to ensure accurate and consistent numerical analysis**.**
    Formula Profit = Sales[Sales]-(Sales[Product Cost]+Sales[Expenses])

- **Profit_Margin Standarized**
  - A calculated column named **Profit_Margin** was created using a DAX formula to compute profit margin as the ratio of **Profit to Selling Price**. The DIVIDE() function was used to handle division safely and avoid errors due to zero or blank values. Additionally, **Clean** and **Trim** transformations were applied to remove unnecessary spaces and hidden characters, and the data type was set to **Fixed Decimal Number** to ensure accurate and consistent numerical analysis.
    Formula Profit_Margin = DIVIDE(Sales[Profit],Sales[Selling Price])

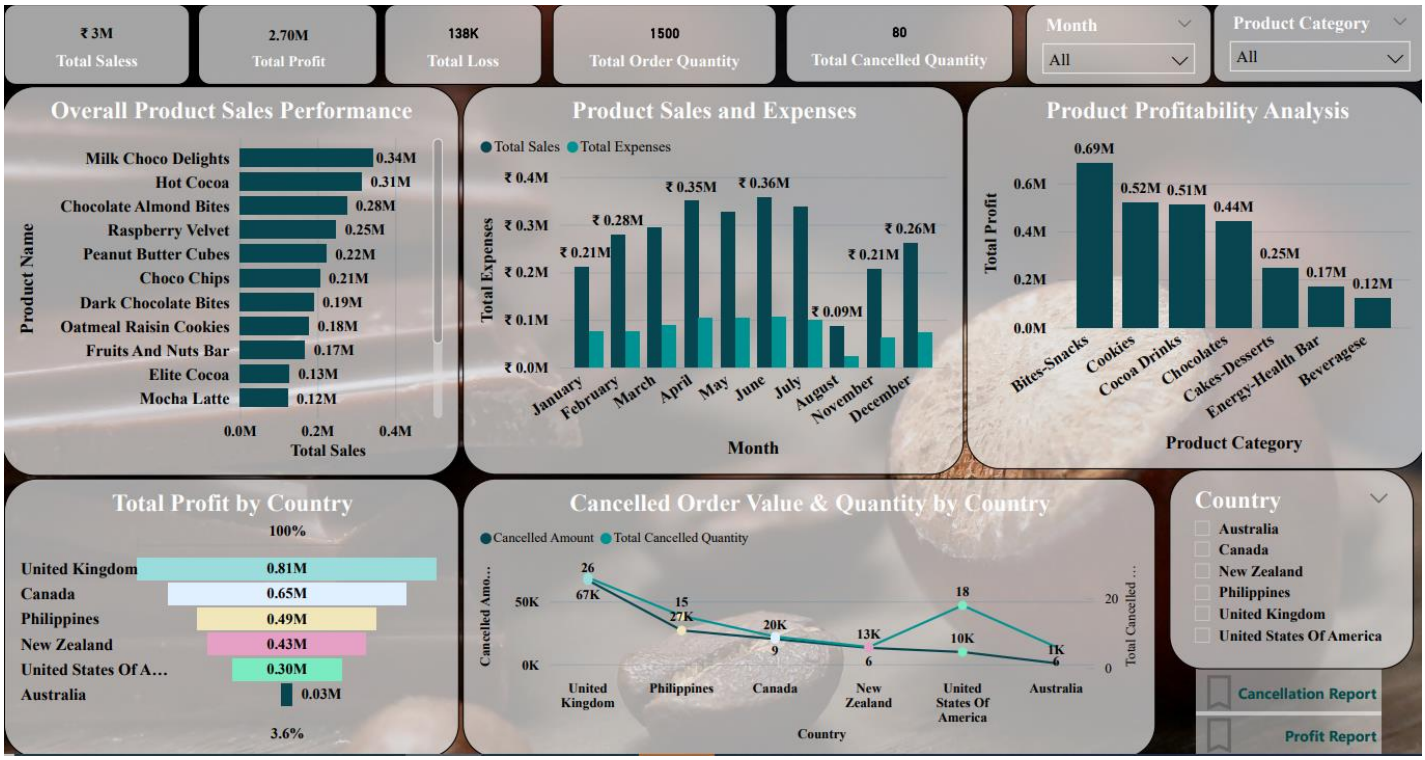
Statistical Summary

Statistical Summary of Sales	
Mean	1805.19
Median	709.20
Mode	132.00
Standard Deviation	2895.80
Sample Variance	8385674.40
Kurtosis	19.24
Skewness	3.67
Range	30501.50
Minimum	1.00
Maximum	30502.50
Sum	2707782.90
Count	1500.00

- Highly Right-Skewed Distribution
  - Mean (1,805.1) >> Median (709.2) >> Mode (132)
  - Skewness: 3.6 (positive, indicating strong right skew)
  - Insight: Most sales transactions are small, but a few very large sales are pulling the average up significantly
- Extreme Variability
  - Standard Deviation (2,895.8) > Mean (1,805.1)
  - Coefficient of Variation: ~160% (extremely high)
  - Insight: Sales are highly inconsistent and unpredictable
- Heavy-Tailed Distribution with Outliers
  - Kurtosis: 19.2 (extremely high; normal = 3)
  - Insight: Many extreme outliers exist - occasional very large sales transactions
- Typical vs. Average Sales
  - Median (709.2) is more representative of "typical" sales
  - Mode (132) shows the most frequent sale amount

- Insight: A typical sale is around $709, but the most common sale is only $132
- Wide Range
  - Range: 1 to 30,502.5 (ratio of 30,502:1)
  - Insight: Massive disparity between smallest and largest sales

Visualization and Insights

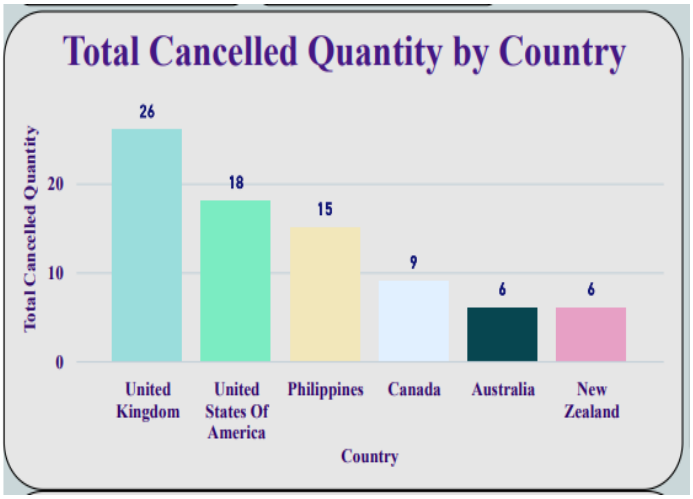


Total sales are approximately ₹3M with 1500 total orders.

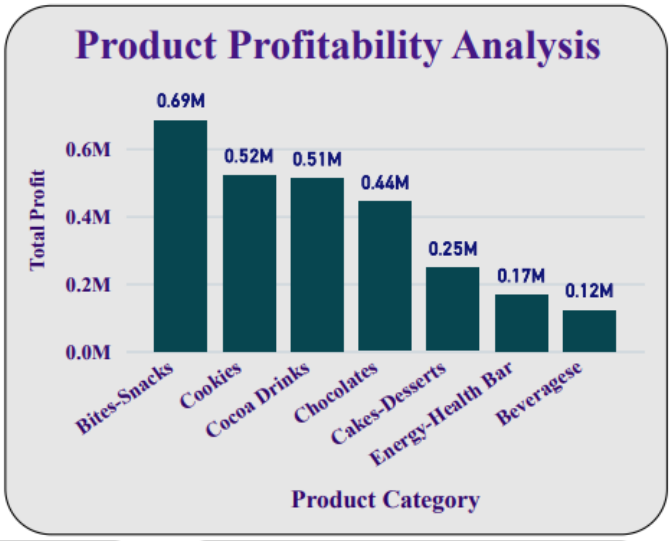
- Top-selling products:
  - Milk Choco Delights
  - Hot Cocoa
  - Chocolate Almond Bites
- Most profitable category: Bites & Snacks, followed by Cookies and Cocoa Drinks.
- Highest profit by country:
  - United Kingdom
  - Canada
  - Philippines

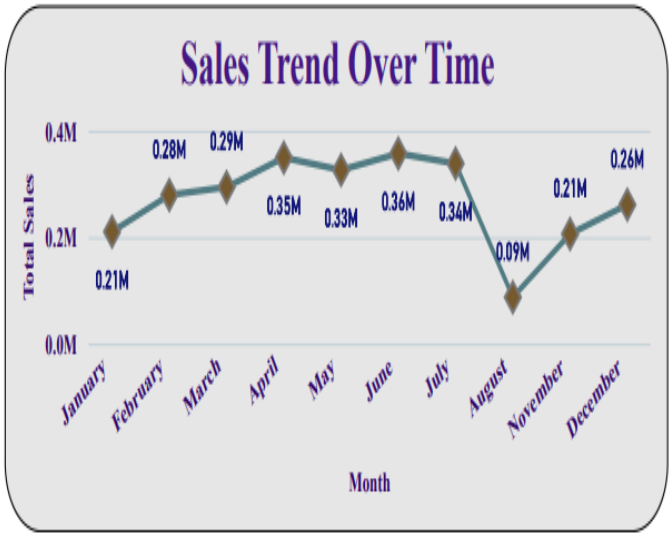
Cancellations

- Total cancelled quantity: 80 orders
- Highest cancellations by:
  - Country: United Kingdom
  - Product: Hot Cocoa, Milk Choco Delights
  - Months: June and July

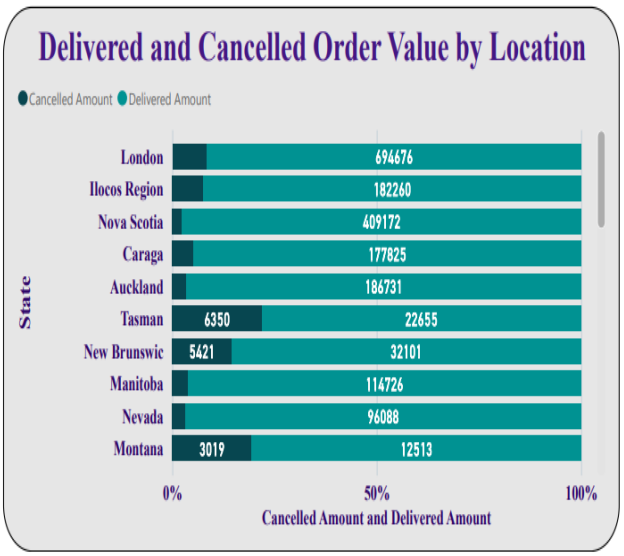


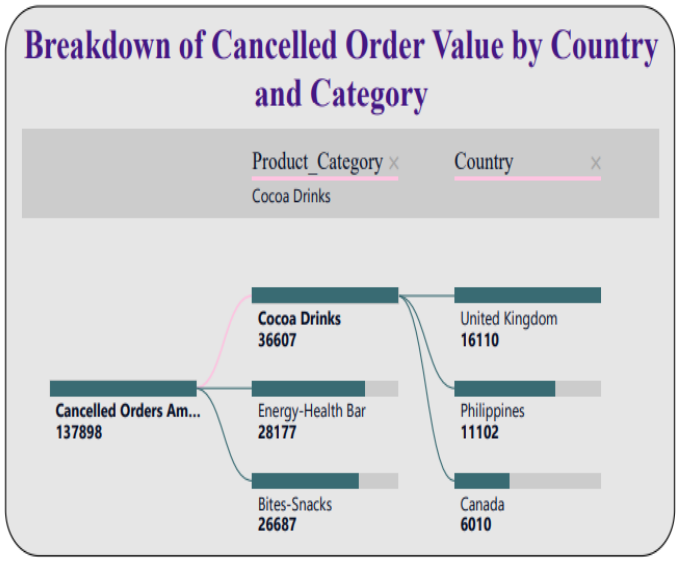
- Higher sales in certain months indicate **seasonal demand patterns**, especially for cocoa-based and chocolate products.
- Increased cancellations in the United Kingdom suggest:
  - Possible **logistics or delivery issues**
  - Potential **inventory shortages**
- Products such as **Hot Cocoa and Milk Choco Delights** show higher cancellation quantities, indicating:
  - Demand exceeding supply
  - Quality or customer satisfaction issues
- Profit dominance of Bites & Snacks suggests:
  - Lower production costs
  - Higher customer preference compared to other categories
- Salespersons with high cancellation counts may be:
  - Serving high-risk regions
  - Handling bulk or complex orders





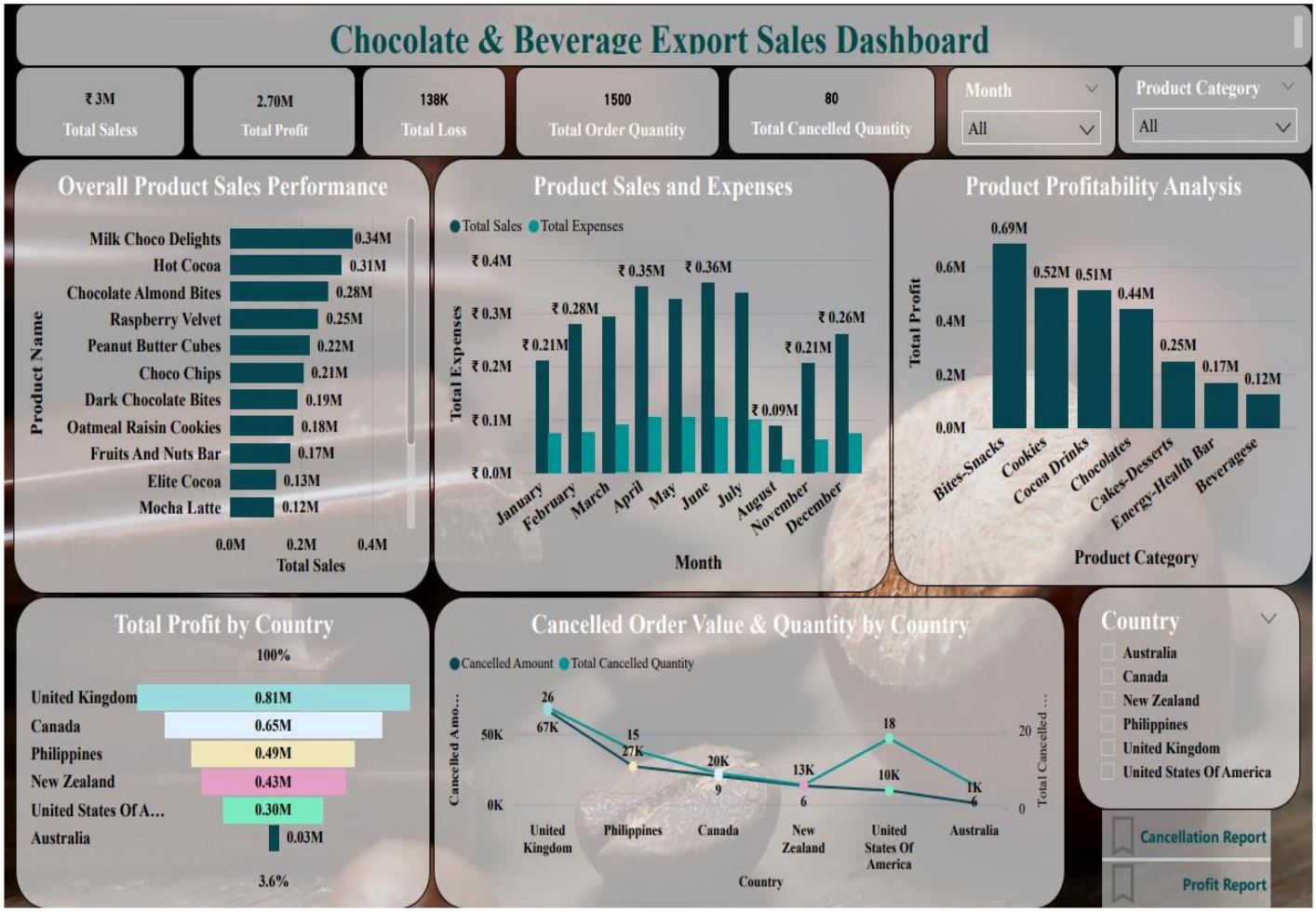
- Sales are expected to increase during peak demand months such as **April, June and July** based on historical trends.
- Cancellation rates may continue to remain higher in:
    - United Kingdom
    - For products like Hot Cocoa
- Product categories such as **Bites & Snacks and Cookies** are likely to remain top profit contributors.
- Countries with strong past performance (UK and Canada) are likely to continue generating higher revenue.
- Salesperson performance trends suggest top performers will continue driving the majority of sales volume.

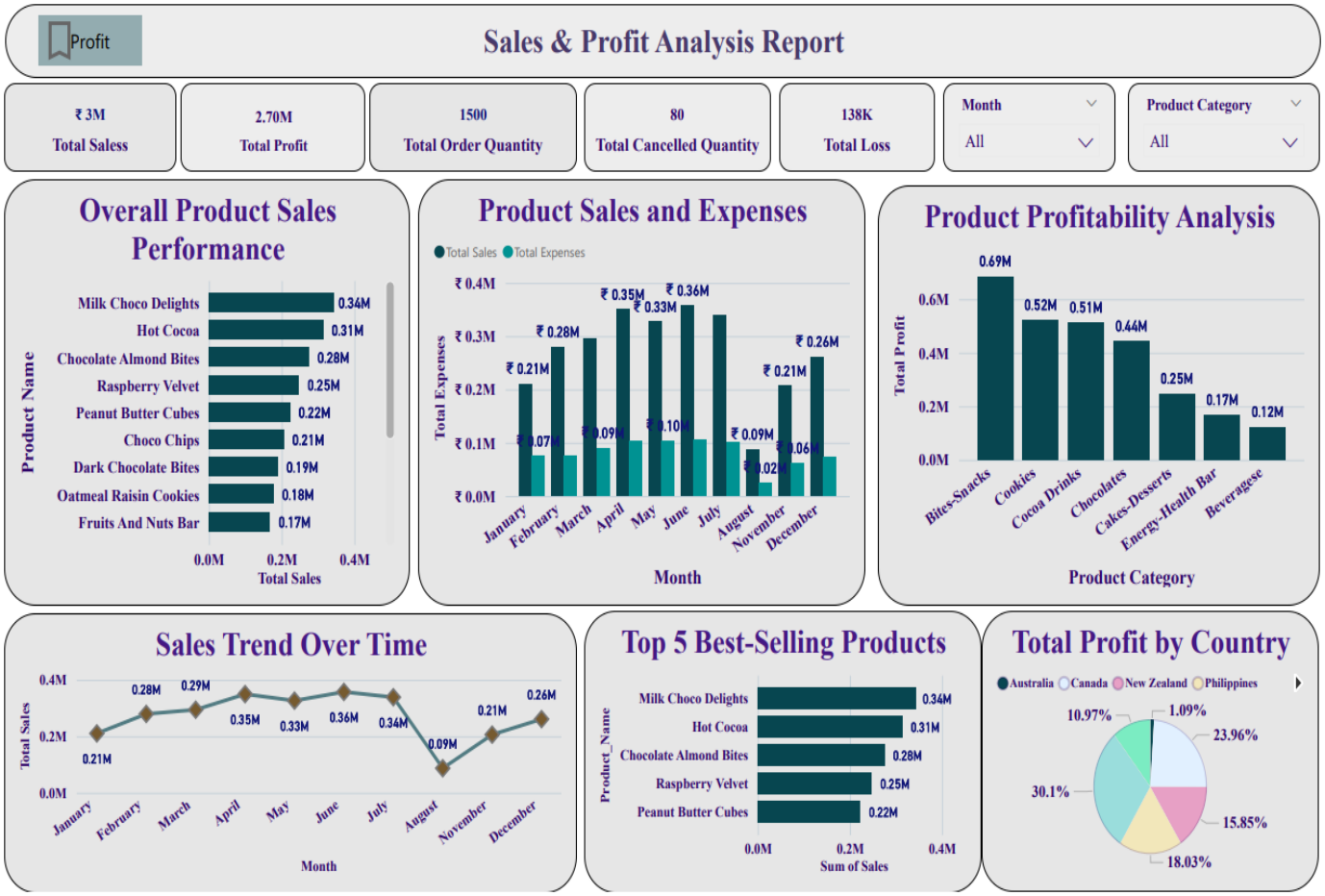


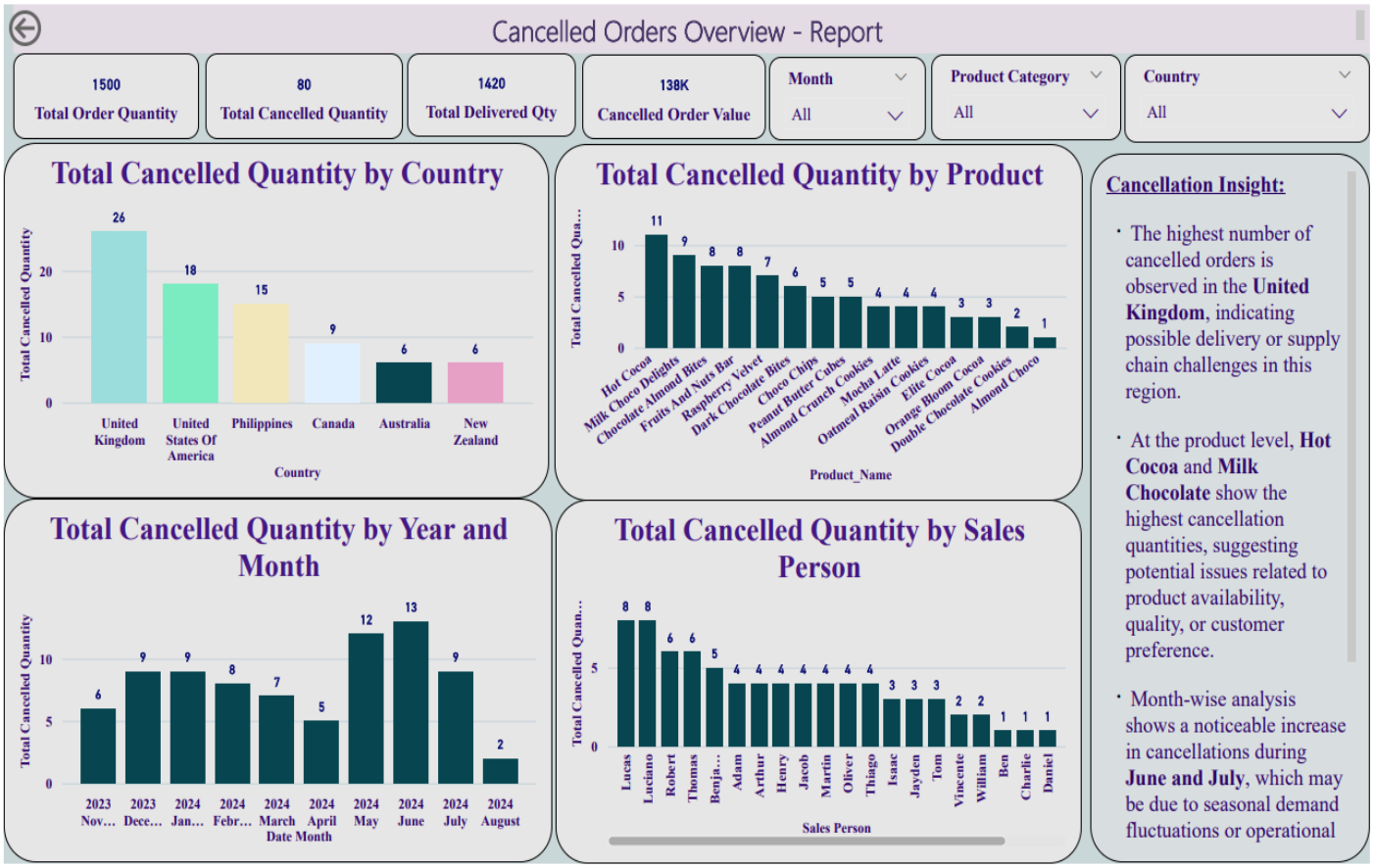


- Increase inventory levels for:
    - Hot Cocoa
    - Milk Choco Delights
    - Chocolate Almond Bites
      during high-demand months.
- Improve supply chain and delivery processes in the **United Kingdom** to reduce cancellation rates.
- Focus marketing and promotional campaigns on:
    - High-margin categories like Bites & Snacks and Cookies.
- Provide performance coaching and support to salespersons with high cancellation rates.
- Introduce demand forecasting models to:
    - Optimize stock planning
    - Reduce overproduction and shortages.
- Implement cancellation root-cause tracking (logistics, quality, delivery delay) for corrective action.

DASHBOARDS AND REPORTS:




Profit
Sales & Profit Analysis Report
₹ 3M
Total Saless
2.70M
Total Profit
1500
Total Order Quantity
80
Total Cancelled Quantity
138K
Total Loss
Month
All
Product Category
All
Overall Product Sales Performance
Product Name
Milk Choco Delights 0.34M
Hot Cocoa 0.31M
Chocolate Almond Bites 0.28M
Raspberry Velvet 0.25M
Peanut Butter Cubes 0.22M
Choco Chips 0.21M
Dark Chocolate Bites 0.19M
Oatmeal Raisin Cookies 0.18M
Fruits And Nuts Bar 0.17M
0.0M 0.2M 0.4M
Total Sales
Product Sales and Expenses
Total Sales Total Expenses
Total Expenses
₹ 0.4M ₹ 0.3M ₹ 0.2M ₹ 0.1M ₹ 0.0M
₹ 0.21M ₹ 0.28M ₹ 0.35M ₹ 0.33M ₹ 0.36M ₹ 0.09M ₹ 0.21M ₹ 0.26M
January February March April May June July August November December
Month
Product Profitability Analysis
Total Profit
0.6M 0.4M 0.2M 0.0M
0.69M 0.52M 0.51M 0.44M 0.25M 0.17M 0.12M
Bites-Snacks Cookies Cocoa Drinks Chocolates Cakes-Desserts Energy-Health Bar Beverages
Product Category
Sales Trend Over Time
Total Sales
0.4M 0.2M 0.0M
0.21M 0.28M 0.29M 0.35M 0.33M 0.36M 0.34M 0.09M 0.21M 0.26M
January February March April May June July August November December
Month
Top 5 Best-Selling Products
Product_Name
Milk Choco Delights 0.34M
Hot Cocoa 0.31M
Chocolate Almond Bites 0.28M
Raspberry Velvet 0.25M
Peanut Butter Cubes 0.22M
0.0M 0.2M 0.4M
Sum of Sales
Total Profit by Country
Australia Canada New Zealand Philippines
10.97% 1.09% 23.96% 30.1% 15.85% 18.03%


Cancelled Orders Overview - Report
1500
Total Order Quantity
80
Total Cancelled Quantity
1420
Total Delivered Qty
138K
Cancelled Order Value
Month
All
Product Category
All
Country
All
Total Cancelled Quantity by Country
Total Cancelled Quantity
26 18 15 9 6 6
United Kingdom United States Of America Philippines Canada Australia New Zealand
Country
Total Cancelled Quantity by Product
Total Cancelled Qua...
11 9 8 8 7 6 5 5 4 4 4 3 3 2 1
Hot Cocoa Milk Choco Delights Chocolate Almond Bites Fruits And Nuts Bar Raspberry Velvet Dark Chocolate Bites Choco Chips Peanut Butter Cubes Almond Crunch Cookies Mocha Latte Oatmeal Raisin Cookies Elite Cocoa Orange Bloom Cocoa Double Chocolate Cookies Almond Choco
Product_Name
Cancellation Insight:
• The highest number of cancelled orders is observed in the United Kingdom, indicating possible delivery or supply chain challenges in this region.
• At the product level, Hot Cocoa and Milk Chocolate show the highest cancellation quantities, suggesting potential issues related to product availability, quality, or customer preference.
• Month-wise analysis shows a noticeable increase in cancellations during June and July, which may be due to seasonal demand fluctuations or operational
Total Cancelled Quantity by Year and Month
Total Cancelled Quantity
6 9 9 8 7 5 12 13 9 2
2023 Nov... 2023 Dece... 2024 Jan... 2024 Febr... 2024 March 2024 April 2024 May 2024 June 2024 July 2024 August
Date Month
Total Cancelled Quantity by Sales Person
Total Cancelled Quan...
8 8 6 6 5 4 4 4 4 4 4 4 3 3 3 2 2 1 1 1
Lucas Luciano Robert Thomas Benja... Adam Arthur Henry Jacob Martin Oliver Thiago Isaac Jayden Tom Vincente William Ben Charlie Daniel
Sales Person

Conclusion:

- The sales dashboard provides a comprehensive view of product performance, profitability, and cancellation patterns across countries and time periods.
- Strong sales contributions from key products such as Milk Choco Delights, Hot Cocoa, and Chocolate Almond Bites indicate consistent customer demand for chocolate-based offerings.
- Product categories like Bites & Snacks and Cookies emerge as the most profitable segments, highlighting opportunities for focused business expansion.
- Regional analysis shows that the United Kingdom and Canada contribute significantly to total profit, while also recording higher cancellation levels, indicating the need for operational improvements in these markets.
- Seasonal variations in sales and cancellations suggest the importance of demand forecasting and inventory planning to avoid stock shortages and order failures.
- Salesperson performance varies across regions, emphasizing the impact of both market conditions and operational efficiency on sales outcomes.
- Overall, the insights derived from the dashboard enable informed decision-making to improve revenue growth, reduce order cancellations, and optimize supply chain and sales strategies.